

Bogdan Mocanu

Engine Programmer

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Netherlands

[Portfolio](#)

Profiles

Github

[OneBogdan01](#)

LinkedIn

[Bogdan Mocanu](#)

Education

Breda University of Applied Sciences

Bachelor of Science

Creative Media and Game Technologies

Aug. 2023 - July 2027

Game Engine and Tools Specialization

Languages

English

Bilingual proficiency

Romanian

Native

Passions

Fitness

Gymnastic Rings, Calisthenics

Baking

Pizza, Pastries, Bread

Video Games

Indie, Grand Strategy, Simulation

About me

I am specialized in engine programming, recently I have explored other areas of software development outside games during my internship, such as 3D reconstruction.

I am most experienced with C++ and I have worked on cross-platform custom engines targeting PC and consoles.

I enjoy learning new things about low-level systems, simulations, engine architecture and optimization techniques by building my own engine. I value being a dependable team member and have collaborated in teams of up to 16 on custom engine projects.

Experience

Radboudumc

Nijmegen

Software Developer Intern

Sept. 2026 - January 2027

I worked on in the 3D lab of the hospital, focusing on 3D reconstruction software.

<https://www.radboudumc.nl/>

Freelance

Remote

Unity Gameplay Programmer

1 JUL 2021 – 1 SEP 2021

In high-school, I become a self-taught gameplay programmer in Unity (C#). I worked on platforms like Fiverr and Upwork during Summer.

Skills

C++

5 years

Vulkan/OpenGL

Using RenderDoc and Nsight

C#

2 years

Shader Programming

Cross Platform development

PS5, Nintendo Switch, PC

Agile Development

Worked teams of 7 to 16 developers

Engine Development

Creating and working with custom frameworks/engines

Build Systems

CMake, MSVC

Game Engines

Unity, Godot, Unreal Engine

Version Control

Git, Github, Perforce, CI/CD

Visual Studio

Debugging

Projects

Thermite | Voxel Raytracing Engine

2026

• **CI/CD pipeline:** hybrid setup with Perforce and GitHub

- **Gameplay systems** (mining, weapons, barge movement)
- **Editor/Engine:** console, build packager, physics layer interface and input module
- Collaborated in a **multidisciplinary team of 16** students

C++ Plugin for Godot | Grand Strategy Map Editor

2025

- Smooth map rendering with **HQX shaders** and **signed distance fields**
- **Export/import** system for **Europa Universalis 4** files
- Custom **Godot inspector** for editing country properties

Hammered | Personal C++ Simulation Engine

2025

- **Multithreaded async logging** system achieving ~11× speedup over single-threaded baseline, profiled with **Tracy**
- Dual-backend **CMake** build system generating both **OpenGL** and **Vulkan** targets from shared codebase
- **GLSL** shaders cross-compiled to **SPIR-V** at build time for both APIs
- Currently restructuring toward modular **ECS** architecture using **flecs**